



Effects of an AI agent-integrated digital game-based learning approach on middle school students' knowledge of internet principles, learning motivation, and classroom engagement

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Received: 9 December 2025 / Accepted: 25 February 2026
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Abstract

Enhancing cybersecurity awareness and digital literacy in the Internet era is increasingly important for young people. Digital games, with their interactivity and immersion, have been applied to the cultivation of cybersecurity awareness among adolescents. However, general digital games lack targeted guidance, making it difficult to sustain deep engagement and limiting academic performance. To this end, this study proposes an AI Agent-Integrated Digital Game Learning Approach (AIDG-approach) that integrates an AI agent-based adaptive feedback system to enhance middle school students' academic performance, learning motivation, and classroom engagement in the Principles of the Internet course. To verify the effectiveness of the approach, 102 students (mean age 12.3 years) from a middle school in eastern China were recruited and assigned to an experimental group ($n=53$) and a control group ($n=49$). The experimental group used a digital game learning approach (AIDG-approach) integrating an AI agent-based adaptive feedback system, and the control group used a conventional digital game learning approach (CDG-approach). The results of the study showed that the experimental group was better than the control group in terms of academic performance and learning motivation, but the difference was not significant; while in terms of classroom engagement, the experimental group had a significant improvement over the control group. The interview results further revealed that the AI agent's personalized feedback feature effectively enhanced students' learning interest and task engagement, but the accuracy of its feedback and task complexity still limited the further improvement of motivation and performance. This study provides empirical evidence for the application of an AI agent-based adaptive gamified learning approach, enriches relevant theoretical frameworks, and offers guidance for design and deployment in educational practice.

Extended author information available on the last page of the article

Keywords AI agent · Digital game-based learning · Learning motivation · Classroom engagement · Academic achievement

Introduction

With the development of digital society, mastering the Principles of Internet not only helps to enhance students' cybersecurity awareness and digital literacy, but also lays the foundation for their future development and further strengthens their participation and innovation ability in the digital society (Brom et al., 2025; Ma et al., 2024). In traditional digital classrooms students are often in a passive state of accepting knowledge and lack sufficient interaction and interest, resulting in low engagement and easy distraction (Demedts et al., 2024; Liu & Shadiev, 2025). Internet knowledge is often abstract and difficult to understand, which further leads to secondary school students being intimidated or even giving up learning in the process of learning the principles of the Internet. Digital game-based learning (DGBL) is a student-centered teaching method that skillfully integrates curriculum objectives and content into digital games to create an interactive and fun learning environment for students (Huang, 2024; Hwang & Chang, 2025; Liat & Hayak, 2025). Empirical evidence has shown that digital games can provide contextualized and socially interactive learning experiences that support the understanding of Internet principles. For example, Jerman Blažič and Jerman Blažič (2022) evaluated 13 cybersecurity digital games in Europe and found that game-based approaches created immersive learning environments that effectively enhanced students' motivation, engagement, and mastery of complex cybersecurity and Internet concepts. Similarly, Zheng et al. (2024) demonstrated that contextualized tasks and immediate feedback in DGBL significantly improved elementary students' netiquette literacy, learning motivation, and engagement.

Although digital game-based learning can provide immersive environments that enhance students' learning motivation and engagement, academic performance is often limited by the lack of timely and personalized support (Ramadhan et al., 2024). To address this limitation, more recent studies have attempted to incorporate feedback mechanisms into digital games. For instance, Mao et al. (2024) explored the effects of dynamic feedback and static feedback on learning outcomes in an educational programming game, and the results showed that static feedback could not adapt to the possible problems that learners might have in different tasks because it did not adjust to the difficulty of the game tasks, which led to reduced learning outcomes and low learning engagement. Likewise, Kuklick et al. (2023) compared the effects of static feedback of different complexity in computerized assessment, and found that simple negative feedback can significantly weaken learners' learning motivation.

Although integrating feedback systems into digital games can help promote learning, most existing systems are based on static rules and standardized feedback mechanisms with limited intelligence, making it difficult to achieve timely and personalized feedback, which limits the learning effect to a certain extent (Huang, 2024). With the maturity of AI agent technology, intelligent personalized feedback for digital games has become possible and has shown initial success in related fields (Hare & Tang, 2023; Tu & Lu, 2025).

Therefore, guided by social constructivist theory, this study developed a digital game integrating an AI agent feedback system and proposed the AI Agent-Integrated Digital Game Learning Approach (AIDG-approach). Subsequently, this approach was applied to the learning of “Principles of the Internet” by an experimental group to examine its impact on learners’ academic achievement, learning motivation, and classroom engagement, in comparison with a control group adopting a conventional digital game-based learning approach (CDG-approach). To evaluate the effectiveness of this intervention, the following research questions are proposed:

- (1) Does AIDG-approach improve learners’ academic performance?
- (2) Does AIDG-approach improve learners’ learning motivation?
- (3) Does AIDG-approach improve learners’ classroom engagement?
- (4) What are the similarities and differences between the two groups of students’ perceptions of digital game-based learning?

Literature review

Learning the principles of the internet

The Internet has become the key infrastructure of the digital age, profoundly affecting the whole society. It is urgent for students to systematically master the knowledge and skills of the Internet, and to improve their ability to retrieve, discriminate and process information based on their understanding of its operating principles, so that they can form rational judgments and make appropriate decisions in the complex network environment (Tüzün et al., 2023). Principles of Internet includes both the basic concepts and technical principles of the Internet, as well as the corresponding operational skills, and is an important knowledge base to support individuals to make safe, effective and responsible use of the Internet environment. Some studies have shown that knowledge of the Principles of the Internet is not only the core foundation of digital citizenship literacy, but also directly related to students’ career and social adaptability (Brom et al., 2025; Chen, 2020). Because knowledge of Internet principles is difficult to acquire through natural exposure, learning about Internet principles in schools remains a key pathway to ensure safe, healthy, and responsible Internet use for middle school students (Choi & Park, 2023). Students often encounter risky behaviors such as password security and privacy exposure when using the Internet, which highlights the urgency and necessity for students to learn knowledge of Internet principles (Hsiang et al., 2024).

Knowledge of Internet principles is a great challenge for middle school students whose cognitive development is still in the transition stage from concrete to formal operations due to its high level of abstraction and conceptual complexity. Brom et al. (2025) showed that middle school students generally have conceptual confusion and cognitive bias in Internet knowledge, especially in the areas of data transmission mechanisms, network infrastructure, and digital privacy, which show obvious Tang and Hare (2023) further pointed out that teachers often rely too much on lectures when teaching abstract knowledge, which not only makes it difficult for students to understand intuitively and operate actively, but also easily leads to their lack of

Learning motivation. Scholars have attempted to introduce new technologies (e.g. VR, digital games) and strategies (e.g. 5E teaching model) to stimulate students' learning motivation and increase classroom engagement to enhance learning performance (Panjaburee et al., 2025). For example, Luse and Rursch (2021) used the ISEAGE Virtual Lab online course as a basis for simulating a real network environment without physical equipment and experimenting with real-world project-based learning. The results of the study show that this model not only significantly promotes the knowledge acquisition of Internet principles and self-efficacy, but also effectively enhances students' hands-on and higher-order thinking skills. In addition, a systematic review by Asadi et al. (2024) demonstrated that simulation-based learning tools (e.g., Cisco Packet Tracer and GNS3) were effective in supporting the teaching of Principles of Internet courses by providing visual, low-cost virtual network environments combined with project-driven and collaborative learning strategies. However, Jerman Blažič and Jerman Blažič (2022) pointed out that traditional learning methods are often difficult to fully stimulate students' interest in learning due to the lack of fun and interactivity, thus limiting the effectiveness of teaching. To deal with this problem, scholars have introduced digital games to enhance students' learning motivation and promote cybersecurity learning (Huang, 2024; Khattib & Alt, 2024). Digital game-based learning is able to transform abstract concepts into actionable and interactive tasks through visual simulations and immersive scenarios, thus compensating for the lack of depth of understanding, engagement, and personalized support in traditional teaching (Tang & Hare, 2023).

Digital game-based learning

Digital game-based learning refers to the process in which learners acquire knowledge and skills by achieving set learning objectives in a virtual situation and rule system constructed by a digital game. In this process, participants are provided with feedback information (e.g., scoring, progress, leveling up, or plot advancement) to guide goal achievement (Clark et al., 2016). Digital game-based learning not only enhances learning motivation (Liu & Hwang, 2024), but also supports knowledge construction and skill development by increasing learning engagement (Gui et al., 2023, 2025). Alt (2023) explored how game design elements can be used to increase students' motivation to engage in mathematics learning activities, and the results showed that problem-based gamification activities were enhancing students' gaming experience and learning motivation. In addition, Chang et al. (2025a, b) proposed a digital game training method based on motivational modeling, which enhanced learners' skill mastery level and engagement motivation through effective interaction with an AI agent. Israel-Fishelson and Hershkovitz (2020) found a significant increase in learner engagement by analyzing student behavioral data in CodeMonkey, a gamified learning platform. Ramos et al. (2023) showed that Network Analyst Board Game (NABG) can simplify the operation of network devices and configurations through abstraction, and allow students to interact and compete in order to deepen their understanding of core computer network concepts and significantly improve learners' knowledge acquisition. A study by Giannakas et al. (2015) showed that digital game learning can significantly improve children's knowledge acquisition and

self-awareness of cybersecurity issues. Gui et al. (2023) two meta-analytic systems evaluated the instructional effectiveness of digital games in STEM learning, and the results of the study showed that digital game-based learning demonstrated significant advantages in science courses, computer courses, and cognitive skill enhancement compared to traditional instruction. Intelligent feedback can significantly enhance learners' academic performance (Chang et al. 2025a, b; Chiu and Hwang 2025; Lin & Hwang, 2025), so some scholars have attempted to integrate feedback systems into digital games to support students' learning more effectively (Ramadhan et al., 2024; Tang & Hare, 2023). For example, Xu et al. (2024) integrated a chatbot into a digital game to significantly enhance students' academic performance and learning motivation in an information technology course. Hooshyar et al. (2016) proposed and validated an Intelligent feedback system incorporating gamified formative assessment in their study, and the results showed that it can significantly enhance beginners' programming learning and problem-solving skills. Although immediate and specific feedback has been shown to help enhance learners' knowledge acquisition, Huang (2024) pointed out that the feedback is difficult to be personalized and real-time due to the intelligence of the existing system, and thus the effect of digital game learning is still limited. From a social constructivist perspective, this limitation may stem from insufficient effective scaffolding for learners in gamified learning contexts (Vygotsky, 1978). A central reason for the mixed and inconsistent evidence regarding the impact of digital gamified learning on academic performance is the difficulty of aligning learners' individualized needs with largely static, one-size-fits-all feedback mechanisms. Therefore, this study incorporates an AI agent based on a large language model into digital game-based learning, which is parameterized and customized to provide timely and personalized feedback to support students' learning of Principles of Internet knowledge.

Digital game-based learning environment integrated with intelligent feedback

Most of the existing digital game learning systems rely on static rules with customized feedback, thus making it difficult to provide timely and personalized feedback, which to a certain extent weakens learning motivation and academic performance (Xu et al., 2025). To cope with this limitation, this study incorporates an AI agent into a digital game learning environment based on Social Constructivist Theory (Vygotsky, 1978), which encourages students to actively construct knowledge in real or simulated contexts. Through role-playing and interacting with the learning materials, NPC (non-player character) and AI agent to complete the tasks, students develop a processual understanding of the Internet principles in exploratory practice, and at the same time promote the mastery and transfer of the knowledge of the Internet principles through collaboration and reflection with the AI agent. To this end, this study adopts the Social Constructivist Theory (Jonassen & Rohrer-Murphy, 1999) as the theoretical basis for designing digital gamification learning activities using the 5E teaching model (Fig. 1). The gamified context guides students to actively construct knowledge and obtain personalized learning support through interaction with

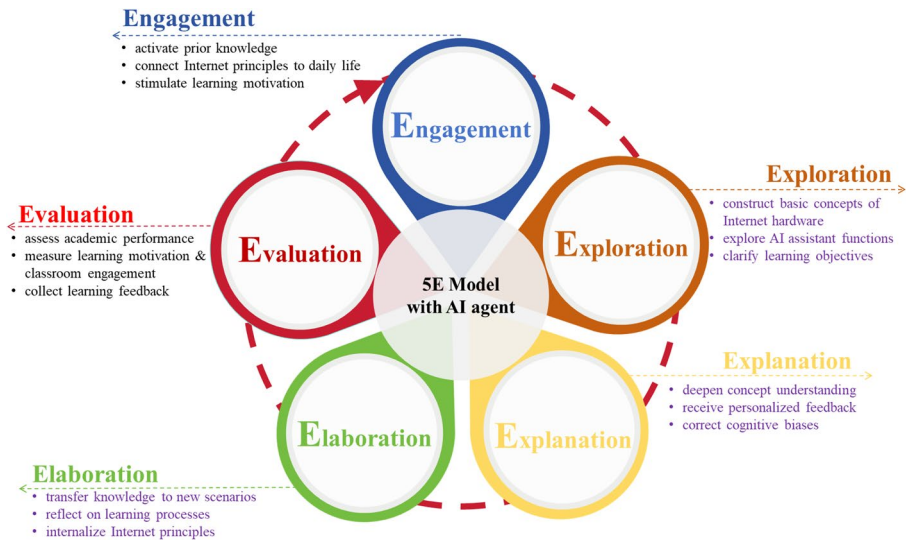


Fig. 1 5E teaching model

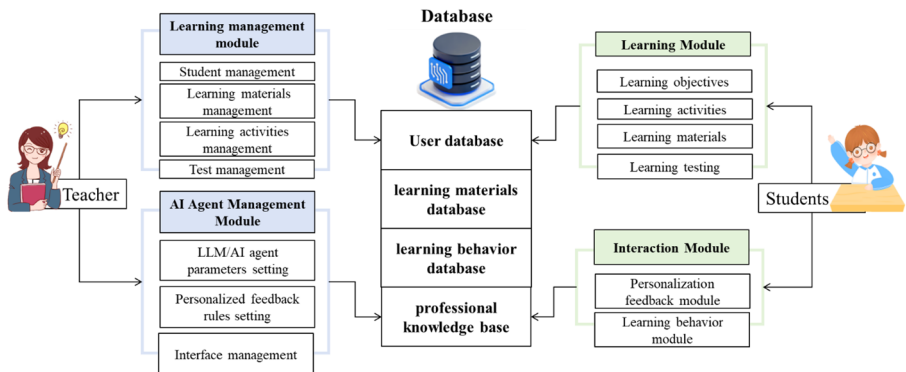


Fig. 2 System architecture

AI intelligence, thus promoting their deep understanding and practical application of Internet principles.

To implement this research design, we constructed a digital game-based learning system integrating an AI agent. The system consists of four core modules: the Teacher Management Module, the Student Learning Module, the AI Agent Module, and the Database Module, and its overall system architecture is shown in Fig. 2.

The teacher's interface includes Learning Management Module and AI agent Management Module. Among them, the Learning Management Module integrates the functions of student management, learning material management, learning activity management and test management, which is used for teachers to upload teaching resources, set learning objectives, design learning tasks and quiz contents; the AI Agent Management Module is responsible for the configuration and optimization

of the intelligences in the system, which mainly includes AI agent parameter setting, personalized feedback rule setting and model interface management. It should be noted that personalized feedback rules do not directly generate feedback content through rule-based triggering. Instead, they function as instructional constraints to guide and limit the construction direction of large language model prompts, ensuring that feedback remains aligned with teaching objectives. To ensure the accuracy and personalization of the feedback, this study developed an AI agent in the experimental group based on the Qianwen3-VL-235B Large Language Model. In this system, the learning feedback received by students is generated in real-time by the Large Language Model based on the learning context and the students' natural language inputs, instead of being called from a predefined static repository of responses. The specific parameters set in the model parameter setting module are: temperature=0.6, which takes into account the logical consistency and creativity; max_tokens=600, which provides a concise and responsive output and reduces the cognitive burden on students; frequency_penalty=0.5, which reduces redundancy and improves accuracy; top_p=0.7, which makes the generated content more consistent. The above parameter settings aim to balance logical consistency with contextual adaptation when generating feedback, thus supporting the model's output of generative cues that are scaffolded and guided, rather than deterministic rule-triggered responses. The personalized feedback rule setting module, based on ensuring the generation of dialogue content that meets the teaching intent, provides students with appropriate, timely and motivational learning support by optimizing the personalized feedback rules. Specifically, these rules are mainly used to qualify the pedagogical function and expression of the feedback prompts (e.g., guided, reflective, or motivational), while the specific textual content of the feedback is dynamically generated by the AI agent in the current learning context. The interface management module, on the other hand, is responsible for connecting the AI agent and Chatbot interfaces to the text comprehension and generation capabilities of the intelligences in the digital game by accessing tools such as the AI agent. Through this interface, the system integrates student input, task context information, and prompt constraints and passes them to the AI agent, resulting in a complete, AI agent-driven feedback generation process.

The student interface includes a Learning Module and an Interaction Module. The Learning Module covers learning objectives, learning activities, learning materials and testing sub-modules, which support students to independently complete knowledge construction and task challenges in a gamified environment; The Interaction Module consists of a Personalized Feedback Module, which provides contextualized feedback and dynamic prompts based on an AI agent, and a Learning Behavior Module, which records students' behavior and interaction logs to provide a data source for learning analysis and personalized support.

The database consists of a user database, a learning material database, a learning behavior database and a professional knowledge base, which are used to store teaching resources, user information and learning behavior data, and to provide data support for the feedback generation and system analysis of AI agents.

Next, the study takes the learning activity 'building a small network' as an example to illustrate the specific process of teaching and learning, as well as the similarities and differences between the experimental group and the control group.

Engagement stage

The learning objective at this stage is to stimulate interest and identify objectives and challenges. Students enter the game as ‘network engineers’ and need to go to ‘Ming’s’ home to help install broadband. At the beginning of the game, there is a fixed NPC close by to guide the development and advancement of the plot, explaining the objectives and challenges of the task (e.g., ‘How can you choose the right equipment so that all the cell phones and computers in Ming’s home can have access to the Internet?’). The link helps students connect learning with daily life experience by creating a real and slightly challenging situation, so that students can intuitively understand the significance of network building in real life, thus stimulating their motivation to learn the knowledge of the principles of the Internet and laying a foundation for subsequent exploration. Specifically, as shown in Fig. 3. The difference between the experimental group and the control group is that the control group completes the task through the textual prompts of the fixed NPCs within the game, while the experimental group provides personalized prompts and feedbacks for the students

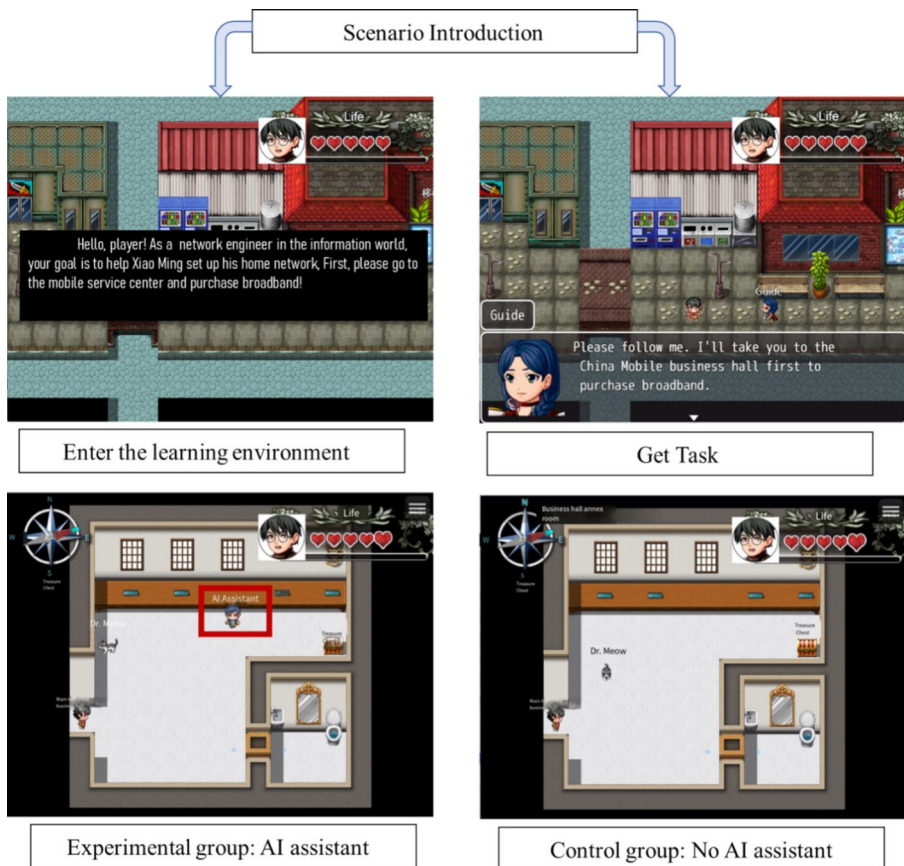


Fig. 3 Situational introduction

who encounter difficulties through the AI Agent integrated in the digital game in addition to the textual prompts of the fixed NPCs.

Exploration stage

During the exploration phase, students begin their initial interactions with the learning environment and gather information on their own. After completing the engagement experience, students are expected to further clarify the objectives and learning content. Both the experimental and control groups are free to explore the digital game and gather information. Then, the two groups of players watch the video ‘Broadband, Router, Wi-Fi, Fiber Optic and Fiber Optic Cats and Their Relationships’ to carry out knowledge learning. This session takes students’ daily experience as an entry point to help them initially understand the basic concepts of network construction, clarify the learning objectives and content, and prepare for the subsequent problem-solving activities. At this stage, the control group can only get limited hints by interacting with fixed NPCs. Unlike the control group, the experimental group can not only interact with fixed NPCs to get hints, but also have open dialogues with the AI agent in the scenario to get more personalized information to help students solve the problem better. Figure 4 shows that the AI agent provides personalized feedback for students.

Explanation stage

The explanation phase is designed to guide students in making connections between the exploration experience and the concept of science. After completing the initial exploration, students acted as ‘network engineers’ to help ‘Ming’ solve the problems encountered in broadband installation. In both the experimental and control groups, students were guided by NPCs to recognize the various types of hardware required

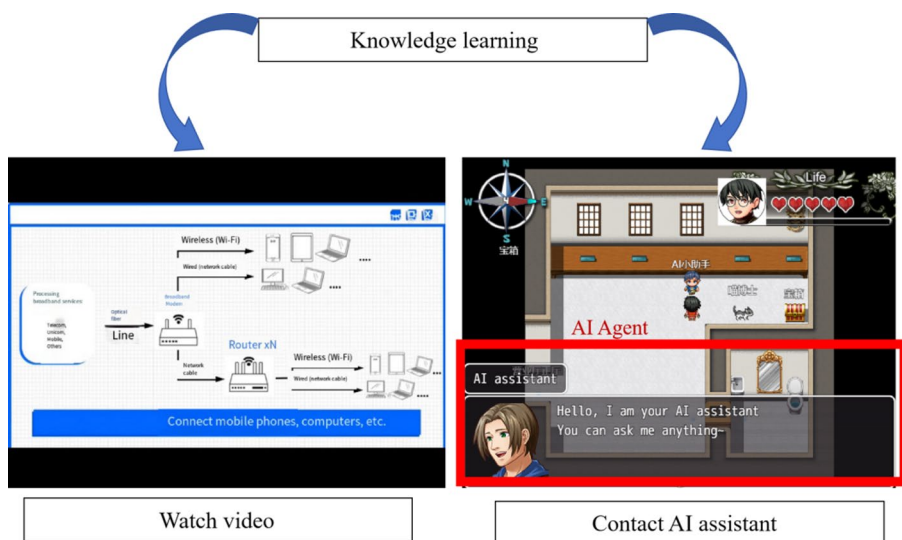


Fig. 4 AI agent offered personalized feedback

for broadband installation and their functions. If they encountered difficulties, the control group sought help from the NPC. The experimental group, on the other hand, in addition to obtaining help from the NPC's prompts, could also interact with the AI agent at any time to get personalized help.

After completing the content learning, both experimental and control groups need to complete the task by answering the test questions proposed by the NPC. When the experimental group completes the test, the system will provide instant feedback on the correctness of the answers: if the answer is correct, the students will not only receive encouragement, but will also be provided with additional explanations to deepen their understanding; if the answer is incorrect, the NPC will explain the reason for the error and encourage the experimental group to look for AI agents to seek help on the map, and at the same time, the player's life value will be reduced by 1 point (with an initial limit of 5 points). When the life value drops to 0, the screen will display 'Game Failed', and players need to watch the learning video again and answer again. After completing the task, the player's desktop will be rewarded with a reward icon and a sound effect, and will be thanked by 'Ming' and other NPCs to reinforce learning motivation. After the control group completes the test, the system will provide feedback on the results, but no encouragement or recommendation of learning materials will be given. Figure 5 shows the interface of the task challenge for students.

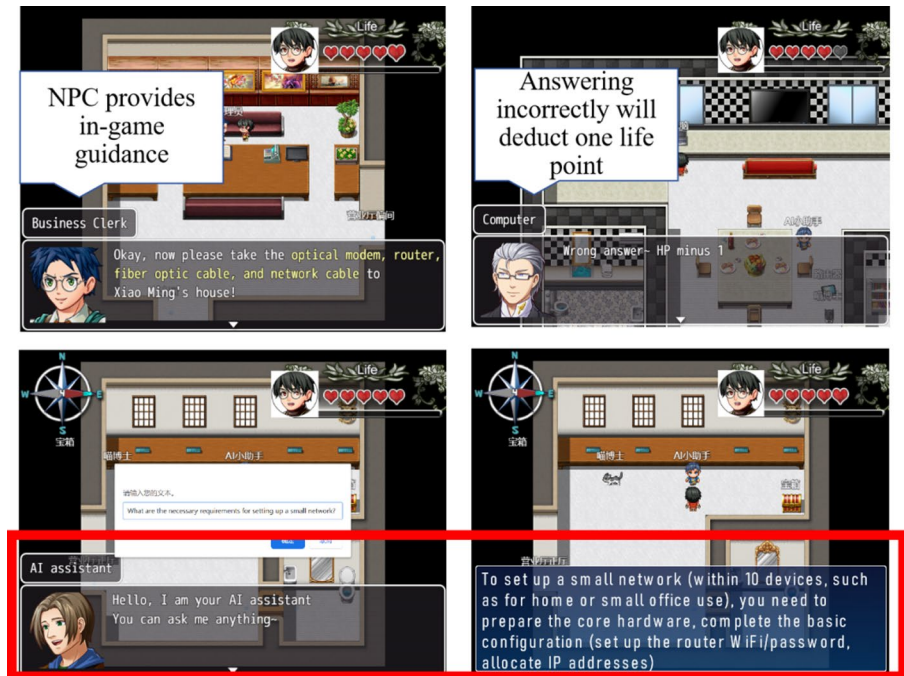


Fig. 5 NPC guides players to seek help from the AI agent

Elaboration stage

During the extension phase, students are encouraged to apply newly acquired knowledge to new complex scenarios to deepen and broaden their understanding. After completing the basic network setup for ‘Xiao Ming’s Home,’ students receive a more challenging task: ‘Design a network for a small office that balances wired connections and wireless coverage while addressing cybersecurity issues.’ Throughout this process, students must integrate knowledge gained in IP address allocation, device selection, and other areas. This stage guides students toward critical thinking and application transfer rather than providing ready-made solutions (see Fig. 6).

At this stage, students in the experimental group can obtain process support through the AI agent, and when they encounter a bottleneck in answering extended questions, they can ask the AI agent questions, at which time the AI agent’s way of answering the questions will guide them to deeper analysis and understanding of the questions, thus promoting the formation of higher-order thinking and migratory understanding. In contrast, students in the control group could only obtain limited guidance through NPCs or system text prompts when performing the same tasks, and lacked immediate, personalized cognitive scaffolding.

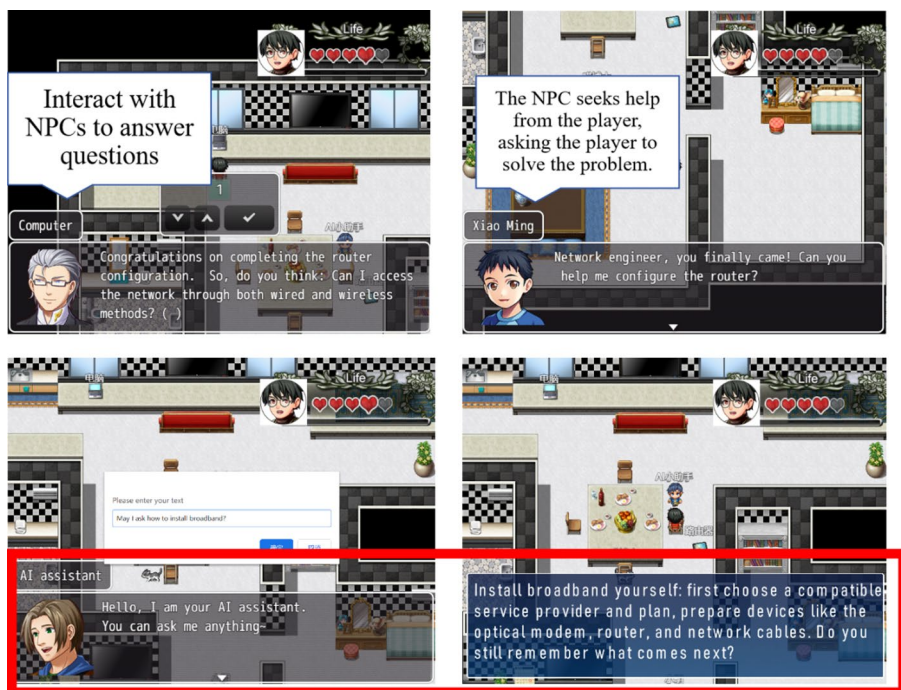


Fig. 6 AI agent guide students to apply and transfer knowledge

information would be kept strictly confidential and that they were free to withdraw at any stage. The study was approved by the Ethics Committee of the researcher's university (approval number: WZU-2024-091).

Experimental procedure

This study utilized a quasi-experimental research design to examine the effectiveness of the AIDG-approach on students' knowledge of Principles of Internet, learning motivation, and course engagement. The experiment was conducted for six weeks, with one lesson per week (45 min). The experimental group used the AIDG-approach, and the control group used the CDG-approach.

In the first week, all learners first completed the Learning motivation questionnaire and the Principles of Internet knowledge test pre-test. Then, both the experimental and control groups spent 20 min familiarizing themselves with the RPG environment and operating the game.

The teaching contents of weeks 2–5 were Internet access, recognizing TCP, IP, IP address and DHCP, network services and Internet application protocols, and building a small network, respectively. During these three weeks, the experimental and control groups accessed the RPG platform and followed the task guidelines to carry out role-playing learning activities related to principles of the internet, which lasted for 40 min. The only difference is that the experimental group's game environment is set up with an AI agent, which can help students answer the encountered questions, and the experimental flow is shown in Fig. 8.

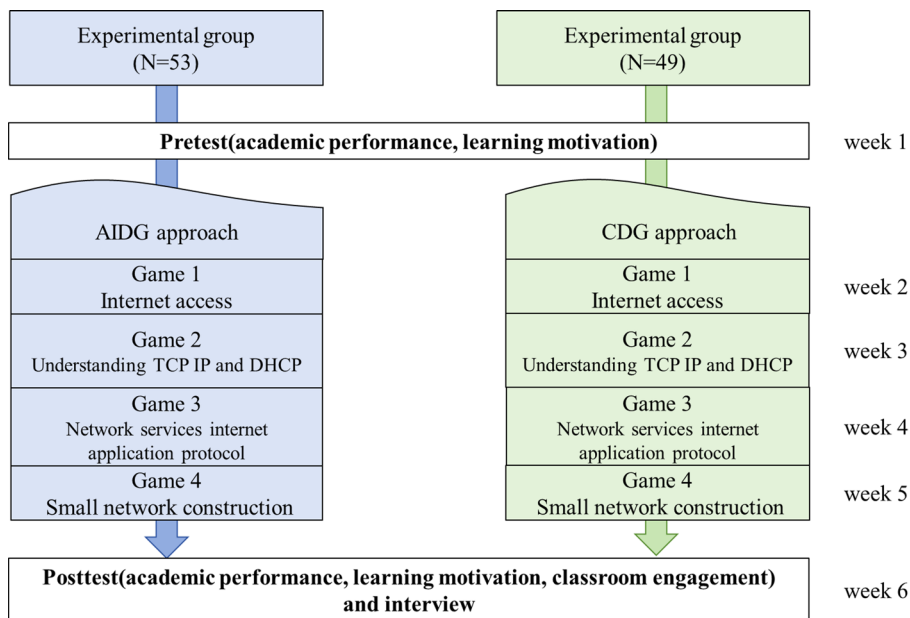


Fig. 8 Experimental procedure

In week 6, all learners completed the posttest of learning motivation and Principles of Internet knowledge test, classroom engagement. Nine students were randomly selected from the experimental group, and nine students were randomly selected from the control group for group interviews, respectively.

Measuring tools

Measurement tools used in this study include Principles of Internet knowledge test questions, learning motivation questionnaire and classroom engagement questionnaire.

The Principles of Internet knowledge test questions are based on questions from proven secondary school information technology exams with known question types and score distributions. Both the pre-test and post-test consisted of 10 multiple-choice questions of 4 points each, 10 judgment questions of 4 points each, and 2 answer questions of 10 points each, for a total score of 100. These questions were assessed by two senior secondary school teachers to determine content correctness and ensure validity. The KR values for the pre-test and post-test were 0.67 and 0.73, respectively, indicating acceptable internal consistency (Cortina, 1993).

Learning motivation questionnaire, which was adopted from Chang et al. (2025a, b), was a 6-question motivation questionnaire with a 5-point Likert scale (1–5), where 1 represents total disagreement and 5 represents total agreement. The Cronbach's alpha value was 0.69, which has acceptable reliability.

Classroom engagement questionnaire was adopted from Elmaadaway (2018) with 25 questions. A 5-point Likert scale (1–5) was used, where 1 represents total disagreement and 5 represents total agreement. For example, I tried to apply what I learned in the classroom. Cronbach's alpha value is 0.7, which has acceptable reliability.

The interview outline used in this study was adapted from Hwang et al.'s (2009) study. On the basis of retaining its core structure and question setting, some of the questioning methods and expressions were moderately adjusted to the context of this study and the characteristics of the subjects to ensure that the questions were more appropriate to the research objectives and the actual situation of the subjects, and the interview and coding contents are as follows:

1. How does this teaching approach differ from the one used in your previous courses?
2. What advantages do you perceive in this learning approach?
3. What the knowledge do you believe you have gained the most from this approach?
4. What aspects of this approach do you think need improvement?
5. Would you like to have the opportunity to learn in this way again in the future? If yes, which subjects do you think are suitable for this approach, and why are these subjects well-suited for it?
6. Would you recommend this learning approach to other students?
7. Would you recommend teachers to adopt this approach in their teaching?

Data collection and analysis

To address research question one, this study collected pre-test and post-test data on academic performance of the experimental and control groups respectively, because the data did not conform to normal distribution, and analyzed them by using the Mann-Whitney U test to examine the pre- and post-intervention students' academic performance of the two groups. and the effect of the intervention on different types of academic performance.

To address research question two, this study collected pre-test and post-test data on learning motivation in the experimental and control groups, respectively, because the data did not conform to normal distribution, Mann-Whitney U-test was used to analyze the effect on the level of motivation in the two groups.

To address research question three, this study collected questionnaire data on classroom engagement in the experimental and control groups because the data did not conform to normal distribution, and analyzed the effect of the intervention on classroom engagement in the two groups through the Mann-Whitney U test.

To address research question four, this study collected learning interviews from students and used qualitative analysis to understand their perceptions of the learning method. In accordance with Creswell's (2013) guidelines, two researchers with extensive experience in the relevant field independently completed the coding, with a Kappa value of 0.82, indicating a high degree of consistency. The researchers first grouped information with similar content into corresponding codes and recorded the frequency of occurrence of each code. They then further integrated the correlated codes to distill themes related to the effects of intelligent feedback and gamified learning. In the course of the coding process, if there was any disagreement, the two researchers would discuss and review the codes again to reach a consensus.

Results

Academic performance

As shown in Table 1, the Mann-Whitney U test results showed that before the intervention, there was no significant difference between the experimental group and the control group in terms of exam paper scores ($U=1241.00$, $p=.70 > .05$, $r=.04$), indicating that the two groups of students had a same level of academic fundamentals before the intervention. After the intervention, the experimental group had a higher mean score on academic achievement than the control group, but the two groups did not reach a significant difference ($U=1172.00$, $p=.40 > .05$, $r=.10$), indicating that the two groups of students had a similar academic performance after the intervention.

Table 1 Mann-Whitney U test results for the academic performance scores of the two groups

Item	Experimental($n=53$)		Control($n=49$)		U	Z	p	r
	M	SD	M	SD				
Pretest	53.00	10.70	52.00	9.82	1241.00	-0.39	0.70	0.04
Posttest	72.00	11.60	70.00	12.47	1172.00	-0.85	0.40	0.10

Table 2 Mann-Whitney *U* test results for the Learning Motivation scores of the two groups

Item	Experimental (<i>n</i> =53)		Control (<i>n</i> =49)		<i>U</i>	<i>Z</i>	<i>p</i>	<i>r</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>				
Pretest	3.67	0.35	3.67	0.47	1195.00	-0.70	0.48	0.08
Posttest	4.00	0.39	3.83	0.53	1201.00	-0.66	0.51	0.08

Table 3 Mann-Whitney *U* test results for the classroom engagement scores of the two groups

Item	Experimental (<i>n</i> =53)		Control (<i>n</i> =49)		<i>U</i>	<i>Z</i>	<i>p</i>	<i>r</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>				
Engagement	4.01	0.43	3.81	0.42	999.00*	-2.01	0.04	0.23

**p* < .05

Overall, the intervention did not lead to a statistically significant improvement in academic performance.

Learning motivation

As shown in Table 2, the Mann-Whitney *U* test showed that there was no significant difference between the experimental group and the control group in terms of learning motivation scores before the intervention ($U=1195.00$, $p = .48 > .05$, $r = .08$), indicating that the two groups of students' learning motivation before the intervention were at a same level. After the intervention, the experimental group had a higher mean score of learning motivation than the control group, but there was no significant difference between the two groups ($U=1201.00$, $p = .51 > .05$, $r = .08$), suggesting that the two groups of students had a similar learning motivation after the intervention. Overall, the intervention did not result in a significant difference in the learning motivation.

Classroom engagement

As shown in Table 3, the results of Mann-Whitney *U* test showed that there was a significant difference ($U=999.00$, $p = .04 < .05$, $r = .23$) between the experimental group ($M=4.01$, $SD=0.43$) and the control group ($M=3.81$, $SD=0.42$) in terms of classroom engagement, indicating that the experimental group students were overall classroom engagement perception was significantly higher than the control group.

Learning reflection from students

This study explored middle school students' perceptions of digital game learning through semi-structured interviews. Specific coding results in Fig. 9 are shown.

In order to better understand the sources of differences in achievement, motivation, and classroom engagement as a result of the intervention, the interview data were thematically coded and analyzed in this study. Students' learning experiences were categorized into 'positive feelings' and 'negative feelings' based on the affective tendencies and semantic features of their statements. This categorization is intended to reveal the structural differences between the 'facilitators' and 'constraints' of digital

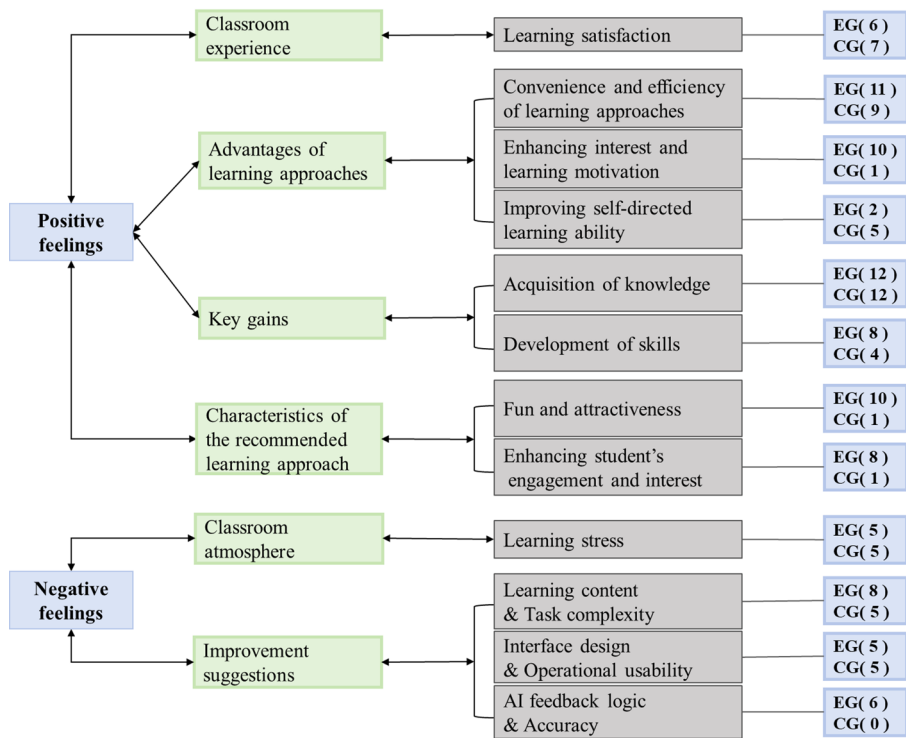


Fig. 9 Coding results of the feeling about Digital game. *Note: EG (Experimental group) and CG (Control group)

game-based learning, and thus explain the changes in learning effects brought about by AI agent intervention more comprehensively.

Positive feelings were coded around classroom experience, advantages of learning approaches, key gains and teaching suggestions. Among them, in terms of ‘advantages of learning approaches’, from the results of ‘convenience and efficiency of learning approaches’ (E = 11, C = 9), the personalized feedback of the AI agent improved the experimental group’s learning experience to a certain extent. The AI agent’s personalized feedback improved the experimental group’s convenience and efficiency of learning, although the magnitude of the difference was limited compared to the control group. The student (E02) mentioned, ‘The AI agent gives me targeted feedback based on my answers so that I can find out what the problem is faster.’ From the results of ‘Learning Interest and Motivation Improvement’ (E = 10, C = 1), the experimental group, with the support of personalized feedback from the AI agent, the students’ interest in learning and learning motivation were significantly improved. The student (E04) mentioned, ‘When I encounter difficulties, I can ask the AI agent at any time, and it always helps me solve the problem, and I begin to enjoy studying this course’. This difference is consistent with the trend of ‘significant increase in classroom engagement’ in the quantitative results, indicating that the AI agent helps students to solve the difficulties encountered in learning by providing dynamic feedback and task guidance, effectively promoting students’ active participation and

learning experience. This indicates that the AI agent helps students solve difficulties in learning by providing dynamic feedback and task guidance, which effectively promotes students' active participation and learning experience. It is worth noting that 'Improvement of independent learning ability' ($E=2$, $C=5$) was mentioned more in the control group. For example, C03 said that although the game was fun, students needed to 'slowly figure it out on their own' when they encountered difficulties in the process of playing the game, reflecting that students needed to have stronger learning ability when there was a lack of feedback from the AI agent. feedback, students need stronger autonomous monitoring and reflection, thus compensating to some extent for the lack of external support. This result supports the social constructivist perspective that 'individuals actively construct knowledge'. In addition, the number of mentions of 'acquisition of knowledge' ($E=12$, $C=12$) in the dimension of 'main gain' was similar in both groups, indicating that the students achieved similar results in knowledge acquisition regardless of whether they received AI assistance or not, which is consistent with the fact that academic performance did not reach a significant level. Acquisition of knowledge' ($E=12$, $C=12$) has similar number of mentions in both groups, indicating that students achieved similar results in knowledge acquisition whether they received AI assistance or not, which is in line with the experimental results that did not reach a significant difference in academic performance. 'Development of skills' ($E=8$, $C=4$), the AI agent's intelligent feedback helps learners by providing immediate personalized feedback and process-oriented guidance. By providing immediate personalized feedback and process-oriented guidance, the AI agent helps learners correct errors and optimize strategies in a timely manner, thus significantly promoting the effective acquisition of skills. In terms of recommended learning methods, the experimental group was significantly higher than the control group. Students in the experimental group generally agreed that incorporating the AI agent digital game learning made learning easier by being able to access help quickly, and were therefore more willing to learn. As student (E04) pointed out, 'When I encounter problems, AI agent can give me answers quickly and help me solve problems, and learning is easier.'

Negative perceptions were mainly focused on 'classroom atmosphere' and 'improvement suggestions'. In the subcategories of 'learning content and task complexity' ($E=8$, $C=5$) and 'interface design challenges' ($E=5$, $C=5$), the number of mentions was similar in both groups, indicating that digital game learning, with or without the use of an AI agent, requires a high level of student experience in terms of task volume and operation. This suggests that, regardless of whether AI agent is used or not, digital game learning puts higher demands on students in terms of task volume and operational experience, which may lead to an increase in cognitive load for some students, thus affecting learning motivation and performance. On the other hand, the 'AI feedback logic and accuracy problem' ($E=6$, $C=0$) was mentioned by the students in the experimental group, which showed that there are still technical limitations of the AI agent in providing personalized feedback, such as the students reported that the accuracy of the AI feedback was insufficient, and even the problem of 'off-topic' or 'logical inconsistency' (E08). These issues undermine the pedagogical mediation effect of AI and explain why AI agents failed to significantly improve academic performance despite enhancing the engagement experience. There was no

difference between the experimental group and the control group in the dimension of ‘classroom climate’, but both groups of students reported higher academic stress.

In summary, digital gamified learning with integrated AI agents makes the learning process more interactive and contextually immersive, stimulates students’ interest and motivation, increases classroom engagement, and facilitates the acquisition of knowledge and skills. However, the instability of AI feedback quality, task complexity, and pacing pressure limit further significant improvements in learning motivation and performance. In other words, although the AI agent was able to enhance interest and engagement in learning, this positive experience did not fully translate into higher learning motivation and academic performance.

Discussion and conclusions

Using a quasi-experimental design, the study systematically collected and analyzed students’ academic performance, motivation level, classroom engagement data and feedback on their learning experience through the Principles of Internet knowledge test, the learning motivation scale, the classroom engagement questionnaire and semi-structured interviews. academic performance, learning motivation, classroom engagement data and learning experience feedback were systematically collected and analyzed. The results showed that the method improved the academic performance and learning motivation of the experimental group, but it did not reach a significant level. In addition, the classroom engagement of the experimental group was significantly higher than that of the control group.

Regarding research question 1, the result indicated that there was no statistically significant difference in academic performance between the two groups, while the experimental group showed a modest increase in academic performance under the AIDG-approach. This finding is not consistent with the mainstream conclusion that Intelligent feedback can effectively enhance academic performance. For example, Tang and Hare (2023) showed that digital games with integrated intelligent feedback significantly improved students’ academic performance compared to digital games without feedback. However, the improvement in academic performance of digital games is not absolute, but depends on multiple factors such as the design of the game to match learner characteristics, task difficulty, feedback mechanisms, and instructional context (Hamari et al., 2023). The non-significant improvement in academic performance in this study may be attributable to limitations in the accuracy of the AI agent’s feedback and a mismatch between the complexity of the game tasks and students’ cognitive load. As Huang (2024) noted, feedback is effective only when it aligns with learners’ cognitive load. McLaren and Nguyen (2023) also noted that simply embedding an AI agent in digital games does not guarantee learning outcomes, which depend on the type of AI agent support and its fit with the instructional design. The results of the interviews further revealed that there are limitations of the AI agent. Some students (e.g., Student E08) reported that the feedback from the AI agent had the problem of lack of accuracy or logical deviation, which made them frustrated with repeated attempts in the learning process. On the other hand, the complexity of the game tasks and the tight pace of the game also became a source of stress

for some students. For example, E02 noted, ‘Although the content is rich, you can’t keep up when the pace is too fast.’ These negative experiences suggest that although digital games and personalized feedback from the AI agent can stimulate interest in the early stages of learning, when the difficulty of the tasks exceeds the students’ ability to handle them, the persistence of motivation is limited, and the positive effect of the AI agent’s feedback is weakened, resulting in a lack of significant academic performance. The positive effect of AI agent-based feedback is also weakened, and academic performance fails to improve significantly. In other words, while AI agent facilitates knowledge construction, its benefits are limited by the balance between task complexity and feedback pacing. If the design fails to fully consider the cognitive capacity and time constraints of learners, excessive social interactions may lead to cognitive overload, thus reducing learning efficiency. In addition, the lack of review and integration by some students at the end of the lesson may also result in immediate gains that do not translate into stable differences in performance. Overall, the effectiveness of gamified learning is not absolute, but depends on whether it is designed to achieve a dynamic match between task difficulty, feedback logic, and learner characteristics (Hamari et al., 2023). In addition, another possible reason why the experimental and control groups did not achieve significant levels of performance is that the tests were predominantly declarative knowledge, focusing on the examination of memorization effects (Perkins & Salomon, 1992).

Regarding research question 2, the result indicated no statistically significant difference in learning motivation between the two groups, while the experimental group showed a modest increase in motivation scores under the AIDG-approach. This result is consistent with the study by Demedts et al. (2024), suggesting that the change in learning motivation was not entirely driven by the technological intervention. The reason for this phenomenon may lie in the fact that the gamification design was not effective in overcoming the limitations of the cognitive difficulty of the subject content and the feedback mechanism, and learners may still experience motivational fluctuations when completing complex tasks. Previous research has pointed out that while the feedback provided by AI enhances immediate assistance and learning path adjustment, gamification design itself may lead to insufficient motivational pull effects if it fails to deeply fit learners’ intrinsic needs (e.g., autonomy, sense of competence) (Chan et al., 2019). The results of the interviews further revealed the intrinsic reasons for this phenomenon. Most of the students in the experimental group reported that the AI agent’s personalized feedback and immediate prompts made the learning process more interactive and supportive, and they felt guided and encouraged in completing the tasks, which led to a higher sense of interest and engagement. These feedbacks suggest that the AI agent does enhance the attractiveness and motivation of the learning context in the short term. However, this motivational enhancement appears to operate more at a contextual, task-driven level—possibly reflecting a novelty effect from initial exposure to the new AIDG-approach—rather than indicating a sustained increase in intrinsic motivation. Some students pointed out that when the feedback from the AI agent was biased or the complexity of the task was too high, their interest in learning would easily decrease. For example, E08 reported that ‘the feedback from the AI agent is sometimes off-topic, and it takes several attempts to get the right hints’, indicating that the unstable logic of the AI agent’s feedback may weaken

the learners' self-confidence. Meanwhile, although the control group students lacked assistance from AI agents, they showed greater autonomy and persistence in the process of independent exploration and problem solving, which balanced the motivation gap to a certain extent. Therefore, the reason for the lack of significant increase in learning motivation is that although the AI agent can stimulate short-term interest at the operational level, it fails to fully satisfy the deep psychological needs of learners such as autonomy and sense of competence (Chan et al., 2019). In environments with complex tasks and fast-paced feedback, learners' cognitive load increases, making it difficult to form stable motivational engagement.

Regarding research question 3, AIDG-approach significantly enhanced classroom engagement in the experimental group, and had a significant advantage in promoting learning interactions and behavioral engagement, a result consistent with Tang and Hare's (2023) findings. The immersive and task-driven characteristics of digital games can stimulate students' active engagement, and the introduction of AI agent further strengthens this engagement mechanism. The immediate feedback and personalized guidance from the AI agent assistant enabled students to receive continuous response and support during the learning process, resulting in a dynamic and interactive learning cycle. Although the findings indicate that the AI agent had only a limited effect on enhancing learning engagement, this effect may have been jointly shaped by other factors. One is the novelty effect: students' initial exposure to an AI agent-based digital game may elicit heightened interest, which tends to fade over time. Another is teacher facilitation (e.g., timely encouragement of exploratory behaviors and light-touch guidance on challenging questions), which may have subtly increased learners' classroom immersion. The interview results further validate this finding. Most of the students in the experimental group mentioned that they not only received immediate help in their communication with the AI agent assistant, but also were motivated to continue exploring. E02 noted, 'The AI agent gives feedback based on my answers, which allows me to find out what the problem is faster.' E10 added, 'The AI agent prompts made me understand the knowledge points faster and more willing to continue the task.' These feedbacks show that the AI agent improves students' concentration and willingness to interact in task execution through timely response and encouragement, which significantly improves classroom engagement. In contrast, although students in the control group also believed that the game helped them learn, they mostly regarded it as a supportive tool and interacted with it less actively (Student C03: 'You have to figure out the problem by yourself slowly, and you need to watch the video or ask the teacher again and again'). This may suggest that, in the absence of immediate external support, control students were forced to rely more on their own efforts to solve problems, a process that may have inadvertently honed their autonomy and persistence.

In summary, traditional teaching strategies (e.g., personalized feedback) in digital game-based learning environments may not always bring significant results (Chan et al., 2019; Demedts et al., 2024). Only by fully integrating the high interactivity and immersion of digital game environments, systematically evaluating and optimizing instructional strategies, and rationally designing cognitive load, task flow, and feedback mechanisms, can we effectively promote student learning. This study provides empirical evidence for an AI agent-based adaptive gamified learning approach,

enriching relevant theoretical frameworks while offering practical guidance and research directions for the design and implementation of digital game-based learning environments (e.g., network principles instruction) across grade levels and disciplines. The results of this study demonstrate the significant impact of a gamified learning approach integrating an AI Agent in promoting classroom engagement, but several limitations remain. First, the pedagogical functions of the AI Agent are still limited. Since the AI agent used in this study is mainly responsible for task prompts and contextual feedback, it lacks in-depth cognitive guidance and support for differentiated learning, which limits its potential to enhance academic performance and learning motivation. In the future, we can optimize the personalized feedback algorithm and adaptive learning strategy of the AI Agent to provide dynamic guidance according to the real-time performance of the learners, so as to enhance its effectiveness in cognitive facilitation. Second, the intervention period in this study was relatively short, and the development of learning ability is a long-term process, so some long-term effects have not yet been able to be fully observed. Future studies may consider a longer intervention design to better track and reveal the dynamic process of children's learning. Third, the small number of participants in this study had a homogenous background, which may affect the general inferences of the findings. Therefore, this study suggests that future studies recruit a larger number of participants from different backgrounds in order to more comprehensively explore the effectiveness of the gamified learning approach in different educational settings, with children of different ages, and in different subject areas. Fourth, this study may also have been influenced by potential confounding factors, such as the novelty effect and teacher facilitation. Future research could adopt more rigorous experimental designs and improved experimental controls to better account for novelty effects and teacher facilitation, thereby enabling clearer attribution of the observed outcomes. Fifth, Feedback bias from AI agents and suboptimal digital gaming experiences may compromise learning outcomes. Consequently, future research should enhance the accuracy of AI agent feedback and optimize the user experience of digital games to minimize the interference of these factors on experimental results.

Competing Interests The authors declare no potential conflict of interest with respect to the research, authorship, and/or publication of this article.

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